

Database Systems I



About Me

Ouldooz Baghban Karimi

Office: SRYE 5103

Web: www.cs.sfu.ca/~oba2/

Education

PhD, Computer Science

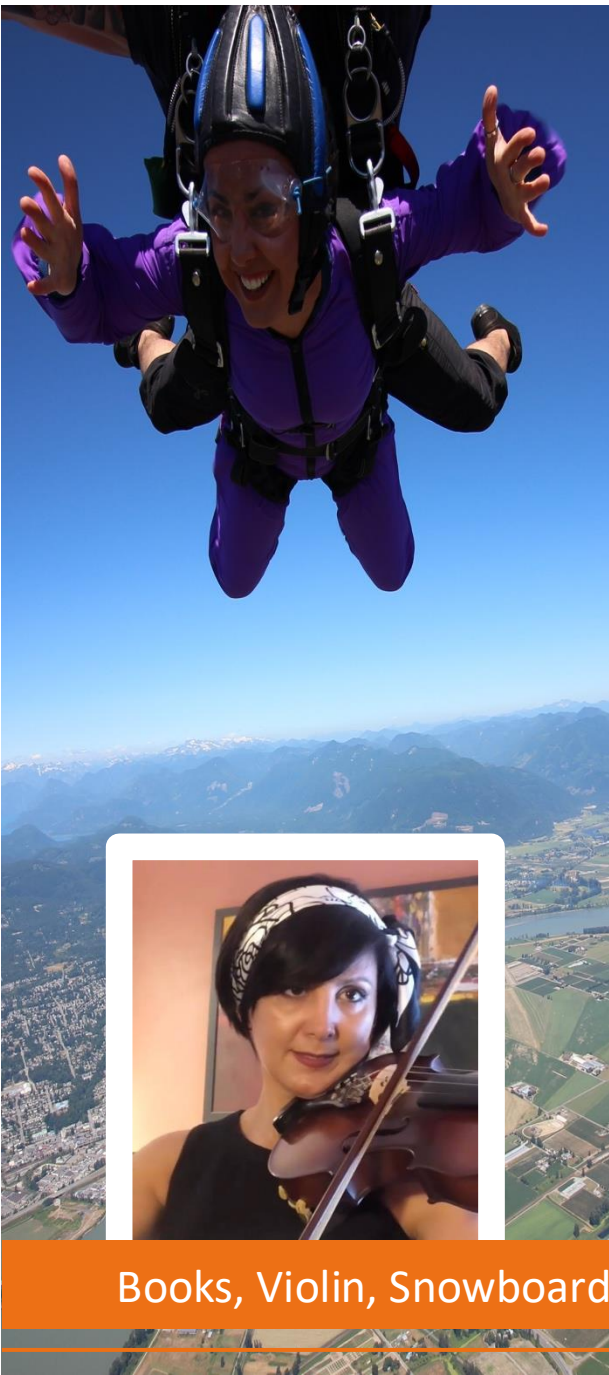
Interests

Networks and Systems

Human-Centric Networks

Diversity and Inclusion





Books, Violin, Snowboarding, Hiking, Travelling, Arts & Crafts

Mengdi Jin (TA)

Education:

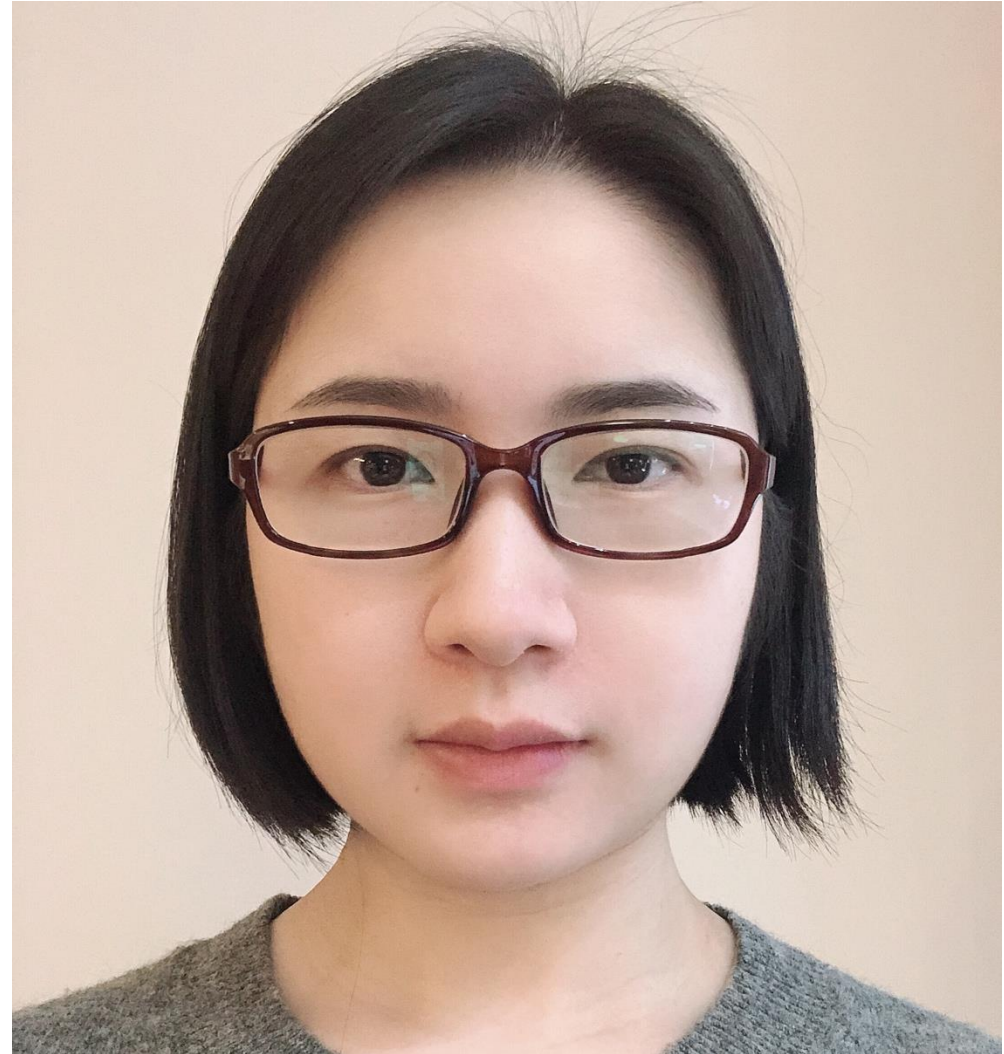
SFU MSc Student

Experience:

NetMedia Lab, working on multimodal AI for ecological monitoring and data collaboration.

Interests:

Networked and multimedia systems
AI for ecological monitoring



Jackson de Faria Junior (TA)

Education:

SFU PhD Student

Experience:

AI

Deep Learning

Fine-tuning LLMs

Research Interests:

Knowledge Transfer for LLMs under
distribution shift



Abhiram Prasanna (TA)

Education:

SFU MSc Student

Experience:

SYNAR lab

Research Interests:

Disaggregated memory systems
RDMA-based database architectures



Topics Overview

- Introduction: Relational Database Definition, History, Applications, and Trends
- Relational Data Model, and Relational Algebra (and Calculus)
- Introduction to SQL: Create, Modify, and Query Relational Databases
- Database Design: Steps and Procedures
- Entity Relationship Model: E/R Diagram Design and Translation to Relational Databases
- Design Theory: Functional Dependencies, Normalization (1NF, 2NF, 3NF, BCNF)
- Constraints and Triggers
- Views and Indexes
- Database Applications
- Query Processing and Optimization
- Transaction Processing: Anomalies, ACID properties, Concurrency Control
- Advanced topics (time permitting)

Course Goals (1)

Gain knowledge about Relational Databases

- What is a database? A collection of files that store related data
- What is a DBMS?
 - DBMS: Database Management System: A piece of software designed to store and manage databases
 - Examples
 - Commercial: Oracle, IBM DB2, Microsoft SQL Server
 - Open source: MySQL (Sun/Oracle), PostgreSQL, SQLite
- Why do we study DBs and DBMSs them?
 - Most popular type of data-intensive system: Relational Databases
 - Many important applications need to manage, update, and query databases (banks, fleet control, search, school system)
 - Data quality, quantity, and timeliness becoming more important with AI & Machine Learning
 - Growth of user and machine generated data

Course Goals (2)

Gain analytical and design skills

- **Design** database systems
 - Data and application needs
- **Analyze** needs of data-intensive applications
 - Data modeling and presentation
 - Challenges of data-intensive systems
 - Reliability
 - Concurrency
 - Performance
 - Accessibility
 - Security

Course Goals (3)

Gain hands-on skills

At the end of this course, be able to:

- **Create** databases
- **Manage** database interactions
- **Create** simple applications using databases

Course Components

- **Lectures:** Recorded (Youtube Videos) + Active Learning (Not recorded)
Recordings are updated every Monday: On normal sessions for the last week. On OAS, for the upcoming week.
- **Interactive Sessions:** Canvas
- **Assignments:** Canvas
- One **Mini-Project** (Group Project: Groups of Two Students, File Submissions on Canvas)
- **Four Quizzes:** In person, on paper
- **Midterm** (Writing Questions)
(90 mins, on paper, in person, Closed-book, one-page two-sided cheat sheet allowed)
- **Final** (Writing Questions)
(120 mins, on paper, in person, Closed-book, one-page two-sided cheat sheet allowed)

Communications

- Announcements
 - Class mailing list
- Active Learning Sessions
 - Assigned discussions
 - Group and class activities
 - Use Active Learning Sessions for additional lecture related questions and discussions
 - Questions and discussions only about the topic (Latest lecture)
- Office Hours (In-Person & Zoom)
 - Use office hours for longer questions

Communications

- Course mailing lists
 - ***cmpt-354-teaching@sfu.ca*** (Teaching Team)
 - ***cmpt-354@sfu.ca***: Announcements (registered automatically)
- Email policy (cmpt-354-teaching)
 - Use email for private/administrative communication
 - Teaching Team Mailing list
 - [354 Urgent] Brief Description
 - [354 Office Appointment Request] Brief Description
 - [354 Request] Brief Description
 - [354 Question] Brief Description

Communications

- Discussion Board (Canvas)
 - Use discussion boards for
 - Quick questions and guidelines on lectures, homework, assignments, etc. (under pinned post) - TA
 - Engage with your classmates in discussions - Students
 - Find study or project groups - Students
- Respectful and Understanding
 - Polite
 - Direct
 - Timely

Office Hours

- **Ouldooz:** Wednesdays, 4-5pm, In-Person, **T9411** (By Appointment)

Office: **SRYE 5103**

Starting the second week of the semester

Book appointments on Canvas Calendar

- **Jackson, Mengdi, and Abhiram:** Drop-in online sessions

Time & Zoom information on Canvas

Starting the second week of the semester

For the By Appointment Office Hours, please book your appointments 24 hours before the session. Please do not book more than two appointments in one day.

Grading

- Assignments & Mini Project **(30%)**
 - Four Assignments (15%)
 - One Mini Project (15%)
 - Late policy (all items): -25% each 24 hours (or less)
- Interactive Sessions & Quizzes **(15%)**
 - Interactive Sessions (5%)
 - Quizzes (10%)
 - Late policy: No late submission accepted
- Midterm & Final **(55%)**
 - Midterm: 20%
 - Final: 35%
 - Late policy: No late submission accepted

Overall passing grade (50%) on the **weighted average of exams (midterm & final)** to obtain a clear pass (C- or better).

Letter Grades

Overall passing grade (50%) on the **weighted average of exams (midterm & final)** to obtain a clear pass (C- or better).

- If the student does not meet the clear pass requirement, but overall grade is >50%, the course grade will be D.
- If the student meets the clear pass requirement, the course grade will be calculated using the overall grade percentage (not only the exams) and the table below:

Letter Grade	From	To
A+	95	100
A	90	94.99
A-	85	89.99
B+	80	84.99
B	75	79.99
B-	70	74.99
C+	65	69.99
C	60	64.99
C-	55	59.99
D	50	54.99
F	0	49.99

Course Resources

Course Web Page: <https://canvas.sfu.ca/courses/13083>

<https://ouldoozbk.github.io/teaching/summer2026/DBS/>

Acronyms used in the schedule:

- **AL:** Active Learning (In-person Problem Solving Session)
- **VL:** Video Lecture (YouTube Videos)
- **IS:** Interactive Session (Online, Hands on)
- **OAS:** Online-only Asynchronous Session (Assigned online videos and activities)
- **SR:** Strongly Recommended Reading
- **OR:** Optional but Recommended Reading
- **OP:** Optional Reading

Textbooks

- **Database Systems: The Complete Book**, 2nd Edition
Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom
Prentice Hall, 2009
- **Database System Concepts**, Seventh Edition
Avi Silberschatz, Henry F. Korth, S. Sudarshan
McGraw-Hill, March 2019
www.db-book.com
- **A First Course in Database Systems**, 3rd Edition
Jeffrey D. Ullman, Jennifer Widom
Pearson, 2007
- **Database Systems: An Application-oriented Approach**, 2nd Edition
Michael Kifer, Arthur J. Bernstein, Philip M. Lewis
Pearson, 2005
- **Database Management Systems**, 3rd Edition
Raghu Ramakrishnan, Johannes Gehrke
McGraw-Hill, 2003